

Jack Gaffney

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SUMMARY

Computer Science graduate from the University of Michigan building real-time AI systems, production backends, and performance-sensitive software in Python, TypeScript, Go, and C++. Strongest when turning ambiguous product ideas into working systems: voice AI pipelines, WebSocket services, LLM workflows, dashboards, APIs, and engines.

EDUCATION

University of Michigan, Ann Arbor, MI

May 2026

Bachelor of Science in Engineering — Computer Science

Coursework: Operating Systems, Embedded Systems, Computer Organization, Artificial Intelligence, Software Engineering, Data Structures & Algorithms, Linear Algebra

TECHNICAL SKILLS

Languages: Python, TypeScript, Go, SQL, C++, Rust, PySpark, Lua

AI Systems: LLM pipelines, real-time voice sessions, transcription/vision, multimodal AI, prompt/config iteration, AI agents, MCP

Backend / Infra: FastAPI, Go/Gin, PostgreSQL, Docker, WebSockets, REST APIs, CI/CD, Linux, Git, AWS Bedrock

Frontend / Product: React, Next.js, TypeScript, operational dashboards, API-driven workflows

EXPERIENCE

Quantum Opus LLC

Plymouth, MI

Full-Stack Software Engineering Intern

Summer 2025

- Built and deployed an internal data platform with Go/Gin, Next.js, PostgreSQL, and Docker Compose, replacing fragmented spreadsheet workflows for engineering and operations teams.
- Designed modular REST endpoints consumed independently by Python data-generation pipelines, reducing redundant cross-service data movement and making integrations more flexible.
- Implemented automated ingestion, validation, and dynamic entity linking to improve data quality and reduce manual cleanup across complex internal datasets.
- Worked directly with technical stakeholders to translate messy operational workflows into scoped requirements, then iterated on backend APIs and UX from real feedback.

Renewit Decking

Charlevoix, MI

Carpenter & Logistics Lead

Summer 2024

- Coordinated logistics and material scheduling for seven distributed crews across a 50-mile radius, reducing downtime across simultaneous job sites.
- Moved from carpentry into logistics and pre-construction planning after identifying recurring workflow bottlenecks, redesigning scheduling processes to improve coordination.

PROJECTS

Tether — Real-Time AI Voice Companion | Go, React, TypeScript, PostgreSQL, AWS Bedrock

2026

- Architected an end-to-end voice AI system with a Go API, WebSocket session lifecycle, PostgreSQL transcript/session persistence, and Amazon Nova model integration for structured voice check-ins.
- Built post-call LLM analysis pipelines that convert conversations into structured insights, follow-up recommendations, and next-call context surfaced through a React/TypeScript dashboard.
- Designed a modular backend across voice sessions, caregiver/patient orchestration, transcripts, analysis jobs, and OpenAPI-documented endpoints with Docker Compose local orchestration.
- Iterated on latency-sensitive voice UX, session state, transcript reliability, and failure handling to make the system usable beyond a demo.

High-Performance Game Engine — C++17, Lua

2025

- Engineered a real-time 2D rendering engine under strict 16ms frame-timing constraints, implementing custom memory allocators to prevent heap fragmentation under sustained load.
- Integrated Lua scripting to expose core C++ systems to higher-level scripts, sustaining 60fps rendering at high actor counts with a software-defined engine architecture.

Vulcan AI — Multimodal Daily Report Generator | Python, FastAPI, PostgreSQL, React, TypeScript

Pi Hacks, Mar 2026

- Built an AI workflow that converts jobsite photos and voice input into branded construction reports using vision analysis, transcription, LLM synthesis, PDF generation, and a React frontend.
- Implemented a FastAPI/PostgreSQL backend for image upload, transcription, report synthesis, and downloadable artifacts, producing finished reports in under 60 seconds.